

MEDICINE LODGE, KS, USA

WMO: 724520

Lat: 37.267N

Lon: 98.548W

Elev: 1535

StdP: 13.90

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03957

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.2	12.8	-2.2	5.2	13.4	1.8	6.4	16.8	29.0	38.6	26.0	38.9	7.7	0	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.8	102.3	72.1	99.3	72.3	96.2	72.3	76.6	91.5	75.3	90.7	74.3	89.4	12.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.6	127.8	82.3	71.3	122.3	80.8	70.2	117.7	80.2	41.1	91.6	39.8	90.4	38.8	89.3	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.6	23.6	20.3	DB	0.9	106.5	5.9	3.2	-3.3	108.8	-6.8	110.7	-10.1	112.5	-14.4	114.8	
			WB	0.1	78.8	5.6	1.3	-4.0	79.7	-7.2	80.5	-10.4	81.2	-14.5	82.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.2	34.6	37.4	47.7	57.3	67.4	77.5	82.2	80.5	72.0	58.9	46.4	35.6	
	DBStd	19.17	10.14	11.55	10.70	9.25	8.57	6.18	5.84	5.90	7.96	9.13	9.51	9.53	
	HDD50	1716	483	375	168	35	2	0	0	0	1	26	175	451	
	HDD65	4322	943	774	541	260	76	2	0	0	27	230	558	911	
	CDD50	4724	5	22	97	256	540	825	997	946	661	302	68	5	
	CDD65	1854	0	0	5	31	149	377	532	481	237	41	1	0	
	CDH74	22095	1	11	92	432	1618	4354	6808	5711	2523	507	38	0	
	CDH80	11455	0	1	17	134	679	2249	3925	3128	1156	162	4	0	
Wind	WSAvg	9.3	8.7	9.5	10.6	11.4	9.8	9.5	8.3	7.9	8.8	9.0	9.0	8.7	
Precipitation	PrecAvg	26.1	0.6	0.9	2.0	2.5	3.5	4.0	2.8	3.3	2.5	2.2	1.7	0.9	
	PrecMax	36.0	2.0	3.0	9.1	7.1	9.3	8.2	8.3	8.2	7.5	7.2	6.1	4.0	
	PrecMin	13.6	0.0	0.0	0.0	0.2	0.2	0.6	0.0	0.4	0.2	0.0	0.0	0.0	
	PrecStd	5.1	0.6	0.9	1.9	1.7	2.3	2.0	1.9	2.4	1.8	1.8	1.6	0.7	
	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.5	75.1	82.2	90.5	97.5	103.2	106.5	105.5	100.6	90.2	78.6	66.8
MCWB			49.7	53.2	58.0	62.3	68.2	71.6	72.1	72.2	70.3	64.9	57.6	51.3	
2%		DB	62.4	68.4	76.2	83.5	92.1	97.9	103.1	101.3	94.8	84.5	73.2	60.7	
		MCWB	46.5	51.1	57.7	61.1	68.5	72.0	72.4	72.2	70.4	64.1	56.2	47.9	
5%		DB	56.8	62.0	71.2	78.9	87.6	94.9	100.0	98.3	91.0	79.9	67.8	56.0	
		MCWB	43.9	48.5	55.6	60.8	68.0	72.0	72.9	72.6	69.9	62.2	53.7	45.9	
10%		DB	51.4	56.2	66.0	74.2	83.1	91.7	96.6	94.6	86.9	75.0	62.8	51.3	
		MCWB	41.4	45.1	53.4	58.9	66.8	71.4	73.1	72.5	68.3	61.2	52.3	42.8	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures		0.4%	WB	52.4	58.3	65.1	68.4	73.6	77.3	78.5	77.8	75.1	71.3	62.2	56.7
			MCDB	63.5	66.5	75.6	79.3	87.0	92.1	92.5	91.9	89.4	80.2	70.1	60.5
	2%	WB	47.9	53.3	61.8	66.2	72.0	75.5	76.9	76.2	73.3	68.4	59.3	50.2	
		MCDB	60.4	63.6	69.1	76.3	85.1	90.2	91.6	91.4	86.7	77.3	66.5	57.8	
	5%	WB	44.7	49.9	58.6	63.9	70.4	74.2	75.4	75.0	71.9	65.9	56.6	46.5	
		MCDB	55.7	62.8	67.0	74.1	82.7	88.8	91.0	90.3	85.0	74.5	64.9	54.9	
	10%	WB	41.9	45.7	54.5	61.3	68.7	72.9	74.3	73.9	70.6	63.0	53.4	43.2	
		MCDB	50.8	55.7	65.0	71.9	79.8	87.1	90.2	88.4	83.2	72.5	62.2	51.1	
	Mean Daily Temperature Range	5% DB	MDBR	23.6	23.8	24.3	24.5	23.3	23.5	23.8	23.4	24.0	24.9	24.7	22.0
			MCDBR	32.9	33.9	32.3	31.4	28.2	27.6	28.5	28.4	27.3	31.2	32.2	29.9
MCWBR			19.7	19.7	17.6	15.6	11.3	8.3	7.2	7.8	9.0	14.5	17.5	18.9	
5% WB		MCDBR	30.4	31.5	25.5	25.6	23.9	23.8	24.0	23.3	23.2	23.5	26.0	27.1	
		MCWBR	19.0	19.3	16.0	15.0	11.6	8.9	8.1	8.3	8.7	13.2	15.9	18.4	
Clear-Sky Solar Irradiance	taub	0.281	0.291	0.333	0.353	0.378	0.403	0.410	0.405	0.370	0.340	0.299	0.278		
	taud	2.514	2.496	2.394	2.376	2.389	2.339	2.347	2.315	2.399	2.460	2.508	2.528		
	Ebn at Noon	292	302	296	294	287	278	275	274	279	279	281	285		
	Edh at Noon	25	29	35	38	38	40	39	40	35	30	25	23		
All-Sky Solar Radiation	RadAvg	841	1077	1420	1768	2003	2234	2196	1963	1634	1212	909	734		
	RadStd	80	143	103	109	167	156	123	110	103	140	101	69		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page